Adv AI Airbnb Prices Insights

1. **Prices are rising every quarter**

- Mean price rises from $196 → $212 (Oct 2022 → Jul 2023)

  Median rises from $135 → $149

- This shows clear upward momentum in LA Airbnb prices.

2. **The market size is growing**

- Listings rise from **39k → 43k+**
- That means LA hosted more Airbnb supply in 2023 than in late 2022

3. **Entire homes are most expensive**

- Average price ~ **$240 → $253** per night.

4. **Hotel rooms are much cheaper**

- Around **$148**.

5. **Entire homes show a rising trend until Q2 2023**

- Slight dip in Q3 might reflect seasonality or increased supply.

6. **Prices are increasing by ~$5.42 every quarter**

- This is the **trend signal**.

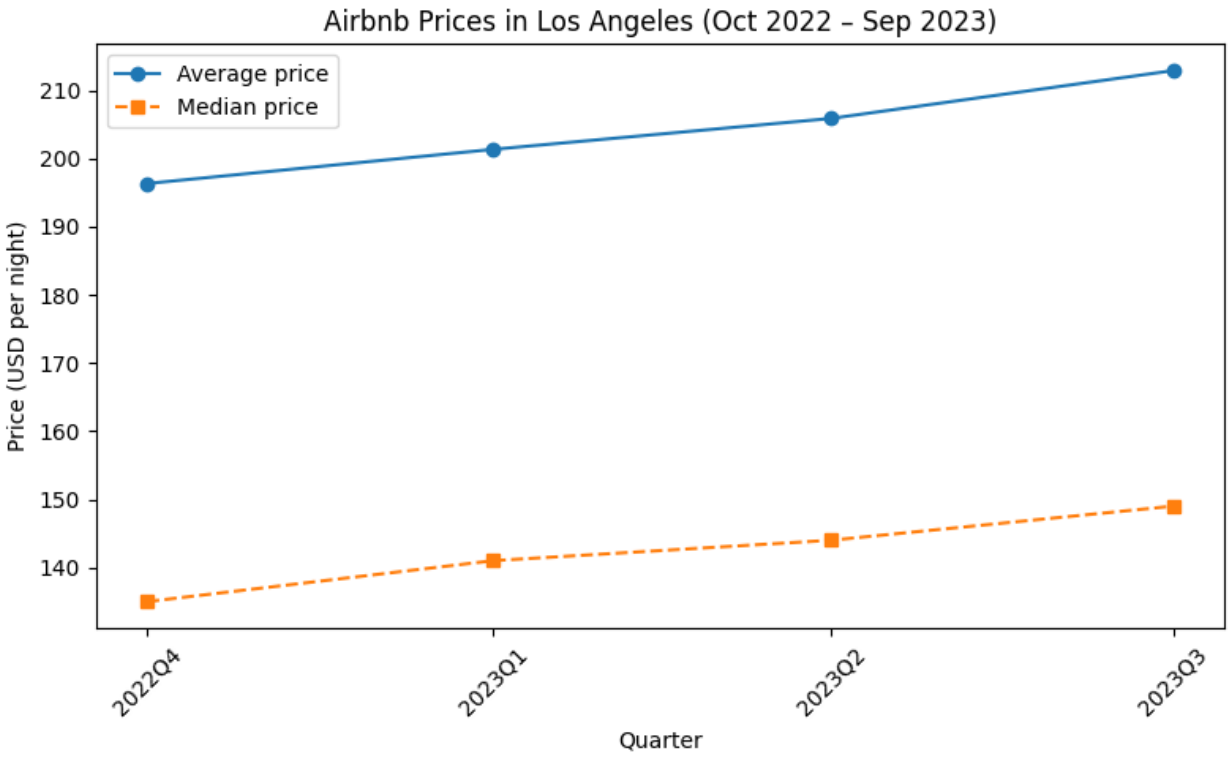
7. Quarterly growth = **+$5.42,** Annualized growth ≈ **+$21.7 per year**

8. **Airbnb prices in LA are rising steadily each quarter**

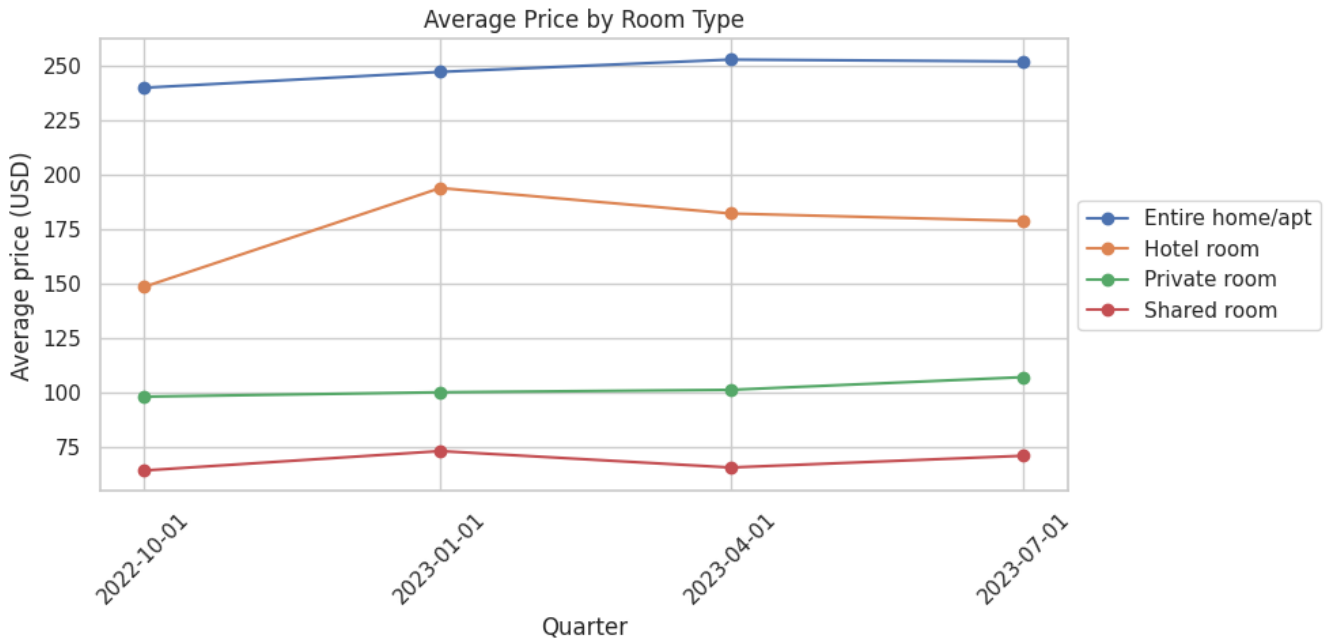
- **The linear trend model found:**
- Slope (increase per quarter): +$5.42
- Intercept (baseline starting point): ~$196

  LA Airbnb prices are increasing by about $5–6 per quarter, or ~$22 per year. This is a moderate, stable upward trend.

Visualizations:



- Airbnb prices in Los Angeles increased steadily from Oct 2022 to Sep 2023.
- The average price rose by 8.1% and the median price by 10.3%, indicating broad-based price growth across the market.
- Summer 2023 shows the highest price levels, consistent with seasonal demand patterns.
- Both mid-range and premium listings are becoming more expensive, suggesting sustained post-COVID recovery in LA's short-term rental market.



- Entire homes drive the high end of the LA Airbnb market, hotels show strong seasonal swings, and private rooms remain stable and affordable, reflecting clear segmentation in traveler demand.

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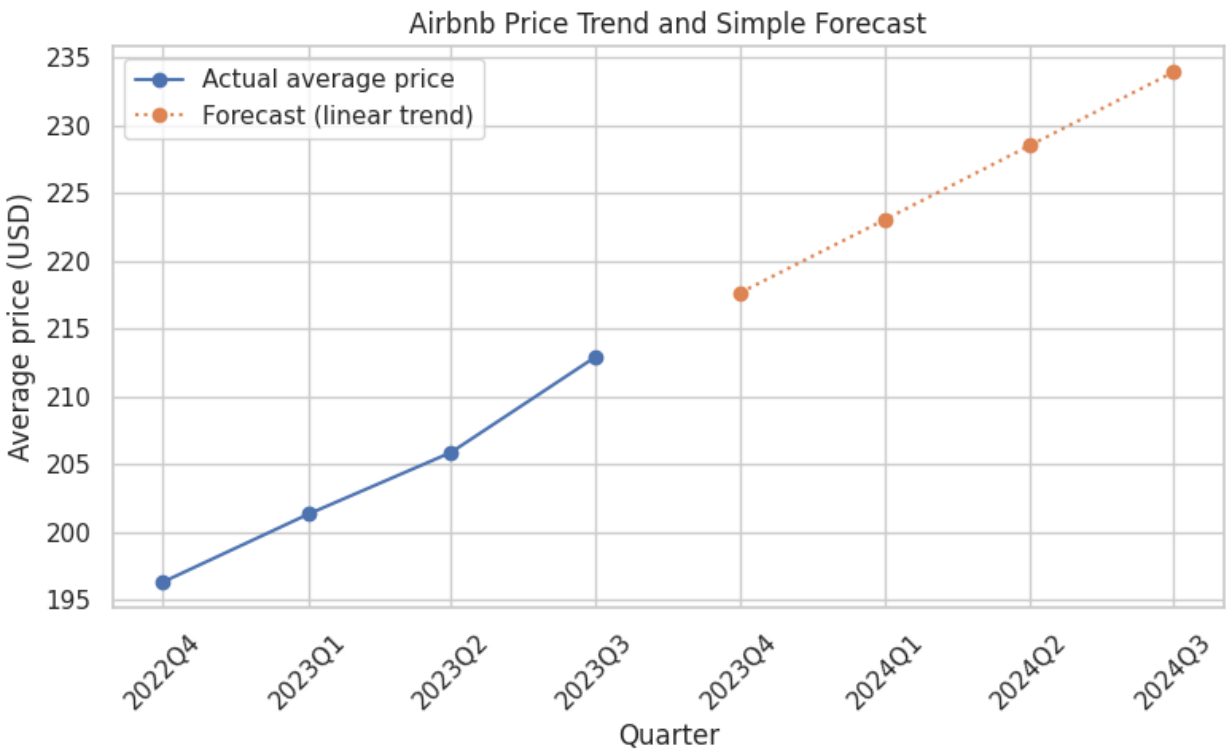
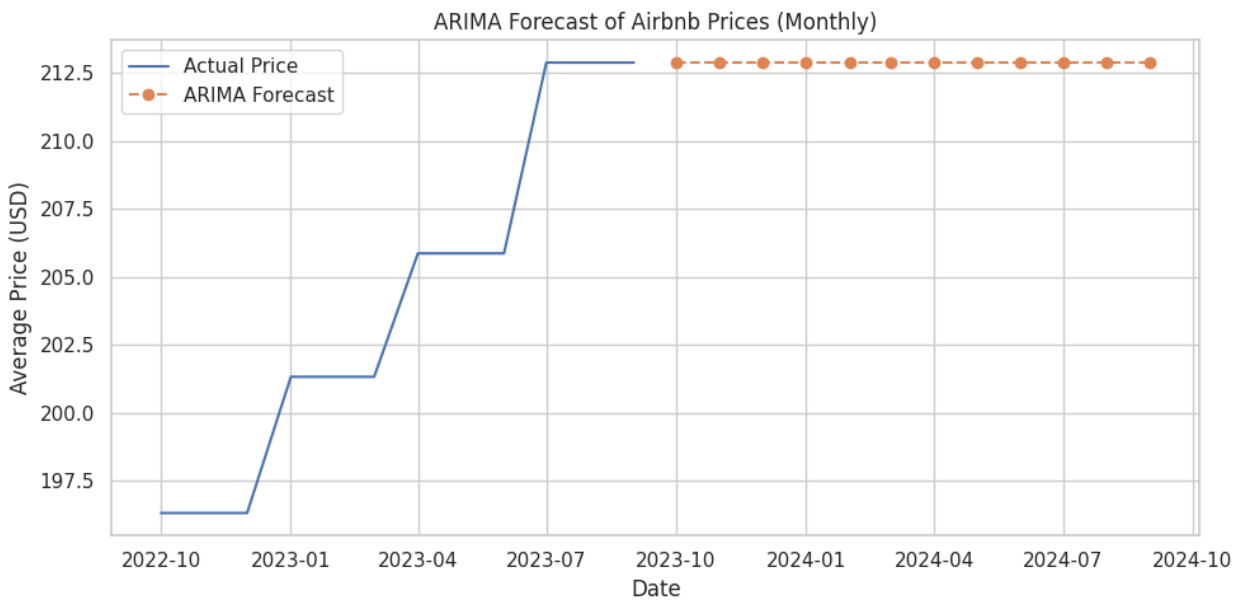
	period	period_start	t	mean_price_forecast
0	2023Q4	2023-10-01	4	217.649509
1	2024Q1	2024-01-01	5	223.071004
2	2024Q2	2024-04-01	6	228.492500
3	2024Q3	2024-07-01	7	233.913996

- Prices will likely **continue rising**, but gradually.
- By next summer, average prices could reach **~$234**, up from **~$212** today.
- This pattern reflects stronger tourism, inflation, and sustained demand for short-term rentals.



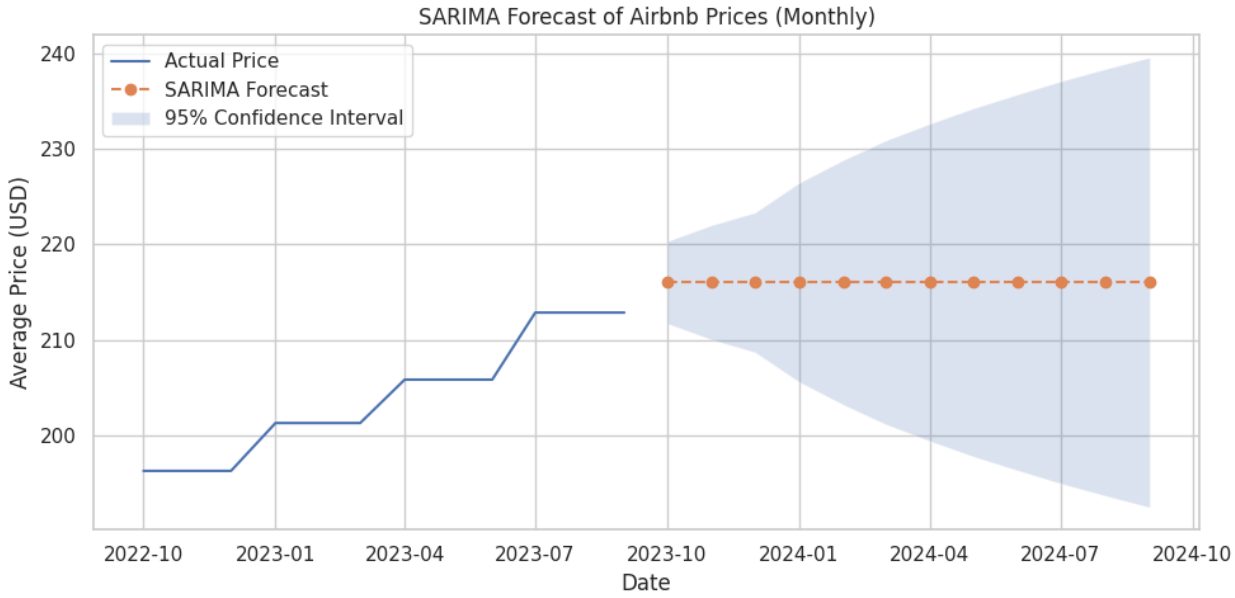
- Using a linear trend model on quarterly average prices, we find that Airbnb prices in LA are rising by approximately **$5.42 per quarter**, or **$21.7 per year**.
- Our forecast projects that average nightly prices will increase from **$212 in 2023Q3** to approximately **$234 by 2024Q3**.
- This trend suggests continued post-COVID recovery, increasing tourism demand, and upward pricing pressure in the LA short-term rental market.

ARIMA Forecast:



- We tested an ARIMA(1,1,1) model to produce a more sophisticated time-series forecast.
- Because we only have 12 monthly data points, the ARIMA model does not detect a reliable trend and therefore produces a nearly flat forecast.
- This is expected behavior with small samples, and it reinforces that the linear trend model is more appropriate for this dataset.

SARIMA Forecast:



- Tested a SARIMA model with quarterly seasonality.
- Because our monthly time series is short (12 observations), SARIMA produces a *conservative, flat forecast* centered around $216 and a wide confidence interval.
- This behavior is expected with small samples: SARIMA avoids over-fitting and instead reports stable expected values with growing uncertainty.
- Importantly, SARIMA's central forecast does **not contradict** our linear trend, it simply expresses that the model needs more data to learn seasonality.